

How to Write Better Research Papers (for CHI)

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ABSTRACT

The challenge of writing good research papers is to communicate a significant and original contribution that benefits human-computer interaction, specifically the CHI community. Researchers have to communicate the validity of their work adequately and do this clearly and concisely to turn a research project into a successful CHI publication. This second online edition of the successful CHI paper writing course offers hands-on advice and more in-depth tutorials on how to write papers with clarity, substance, and style. The instructor has structured it into four online units focusing on the structure and style of the abstract and introduction of CHI papers.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *HCI design and evaluation methods*; *Systems and tools for interaction design*.

KEYWORDS

Writing, Reviewing, Research Methods, Style, Clarity

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1 INTRODUCTION

Writing research papers that communicate our research contribution clearly and allow reviewers to assess the validity of our methodologies with ease is difficult but at the heart of our craft as CHI researchers. Knowing what reviewers are looking for in a paper helps us write and structure papers more clearly. Yet, skillful writing sometimes seems ephemeral to us when trying to structure our research ideas around what we perceive as the CHI community's demands. In the last five years of teaching this course, I have moved from in-person writing seminars to online presentations and guided exercises to a now fully-online class that teaches research authors the basic principles of good writing. I have reworked the

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course structure with even more focus on writing exercises and to allow for guided discussion around an online presentation of the seven common CHI contributions[13]. While some helpful work exists online¹ with lots of writing and research structure advice for CHI authors [1, 3, 4], it has only recently been synthesized into this hands-on course format [5–9].

2 BENEFITS TO PARTICIPANTS

Students will learn the practical writing skills that enable them to “trim the fat” in their writing and focus on bringing the essential information about their research across in the first course unit (some of the advice being taken from essential style guides [2, 11, 12, 14]). They will also learn to dissect a CHI paper into its units and map those units to their research structure. In the second course unit, we will focus on writing the introduction and abstract. In an online exercise, they will learn how to structure their CHI paper around a research narrative that focuses on solving the main problem and outlining a strong contribution using exploratory learning. We will outline CHI papers by sections and draft in the bullet points for each section that will eventually lead to the full paper. In the third course unit, we focus on a discussion of the drafts from the second unit and a review of recent CHI writing data. This unit is focused on online discussion. The fourth unit will focus on polishing the research paper. By following a clear structure and focusing on lean writing, participants will learn what reviewers are looking for and how to signpost this information to make papers more attractive to read.

2.1 Common Writing Tips for CHI

- (1) Start with an outline when writing your paper. Work out the argument first. What is the progression of ideas that you are going to convey to the reader?
- (2) Fall in love with your favourite style guide [2, 11, 12, 14]. Little things like grammatical errors, poor use of parallel structure, and sentences that do not agree in number or sense from one end of the sentence to the other just provide little bumps along the way of the review where a reviewer now has to work harder to find out what you did or why your contribution is important. Any time you make a reviewer work hard, you lose ground, and you lose marks toward their final impression (and score) on your paper.

Participants will leave this course with improved writing skills, which they will have exercised during the course and also a wealth

¹<https://chi2020.acm.org/authors/papers/guide-to-a-successful-submission/>

of knowledge about CHI paper structure, style, and content, which will be expanded online

3 INTENDED AUDIENCE

This course introduces writing principles, including the newest meta findings of CHI papers (from an alt.CHI paper [10]), to a mainly junior audience. However, this does not mean that this course is not useful for senior CHI researchers, but junior researchers are the primary target audience. Thus, this course is beneficial for young researchers, ranging from graduate students to postdocs and junior faculty. The expectation for the course audiences is that people have at least tried to submit a paper to CHI before (not necessarily that they have had one accepted). They are familiar with basic PCS terminology and the CHI conference concept (and CHI research in general).

4 PREREQUISITES

There are no prerequisites for this course other than visiting the online course materials² before the conference to familiarize oneself with some of the course concepts and to listen to the interviews³. The participants are expected to have a computer with a stable Internet connection, online video software with microphone and webcam available.

5 CONTENT

The course is structured into four online units (see Table 1), the first one on research structure, the second one on the Abstract and the introduction, the third one on a hands-on writing tutorial for CHI papers, the fourth one on polishing. At the start of the first unit, the participants are introduced to the course instructor and the course goals:

- Become a better writer by learning how to avoid unnecessary words and give each sentence a strong meaning
- Understand the goals of CHI research and what makes a strong CHI contribution
- Learn how to structure your paper around a compelling research narrative to emphasize your research problem and solution as main drivers

5.1 Unit 1: Paper Structure and Style

This lecture introduces the positioning of research papers and how to narrow the research problem space toward a unit that can be presented at CHI. It will give the participants an overview of what reviewers are looking for in a CHI paper. I will showcase some video interviews that I did with senior CHI researchers and discuss findings from my podcast. The practical activity here is a post-it note exercise to pinpoint important sections of an example CHI paper.

5.2 Unit 2: Abstract and Intro

In many of the interviews that I have conducted with senior CHI researchers, the abstract and introduction have been mentioned as

²<http://writing.chicourse.com>

³<https://anchor.fm/how-to-write-chi-papers>

Unit 1 Structure & Style	
10 mins	Introduction Video & Goals
40 mins	Structuring your research paper - Interactive Online Whiteboard Exercise
30 mins	Dissecting a CHI paper - Group Call
Unit 2 Abstract & Intro	
10 mins	Abstract and Introduction structure video
70 mins	Guided Online hands-on exercise for Abstract and Introduction Writing
Unit 3 Fundamental Discussion	
40 mins	Presentation and discussion of abstracts and introductions from previous unit
10 mins	Review of recent alt.CHI [10] paper
30 mins	Discussion of the recent data set
Unit 4 Polishing	
10 mins	Polishing video
30 mins	Polishing exercise with the existing drafts
40 mins	Discussion the polishing attempts in an online call

Table 1: Structure of four online course units.

the most important structural parts of a CHI paper. This exercise will give the participants work materials for building clear abstracts and introduction to their CHI research work that helps to structure their introduction section.

5.3 Unit 3: Discussion of Own Work and Current Research

Building on the abstracts and introductions that we have built in the previous exercise, we will now discuss the abstracts and introductions that we have written and pinpoint their strengths and weaknesses. We will then review the best practices outlined in the recent alt.CHI [10] paper about writing CHI papers.

5.4 Unit 4: Polishing the Paper Drafts

Often the hardest work is at the end of the writing process when a paper needs to be polished to have a frictionless journey through committee discussions. We will review tips and tricks for making the paper presentation smoother and helping authors convey their information most effectively.

6 PRACTICAL WORK

All four units feature bits of practical work, ranging from guided writing with me rotating around tables and giving people feedback on their writing and online post-it notes exercises (e.g., using Miro online software) to pinpoint important areas of an existing CHI paper. The course work aims to strike a balance between practical writing and the online discussion of participants.

7 INSTRUCTOR BACKGROUND

Lennart E. Nacke, Ph.D., is an Associate Professor for Human-Computer Interaction and Game Design at the University of Waterloo. He has taught the previous four iterations of this course at CHI since 2017. He runs the successful *How to Write CHI Papers* podcast and an interview series about writing CHI papers on YouTube. He has served on the SIGCHI program and steering committees and taught University graduate classes on HCI research methods. Dr.

Nacke has co-organized many workshops for CHI over the past five years; he also chaired the CHI PLAY 2014 and Gamification 2013 conferences, served as technical program co-chair for CHI PLAY 2015 and CHI Games and Play subcommittee co-chair for CHI 2017, INTERACT 2019 Full Papers Co-Chair and was the chair of the CHI PLAY steering committee until 2018. He has also reviewed hundreds of papers and gotten lots of his own submissions rejected from CHI.

8 RESOURCES

The following resources are available to participants.

- How to Write CHI Papers Website
- How to Write CHI Papers Podcast
- How to Write CHI Papers Mailing List
- How to Write CHI Papers Interview Series

9 ACCESSIBILITY

To make everything accessible, all materials will be provided as recordings and transcripts, so that participants can follow along. For all website materials, the instructor will ensure to provide meaningful ALT text for images and colourblind-safe colour schemes. Recorded Online Videos will be subtitled.

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